Employee Attrition Prediction Report

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Dataset Overview:

- Total records: 1470
- Attrition rate: 16.12%
- Features used: 30

Challenges Faced:

- Imbalanced target variable with low attrition rate.
- Need to compare multiple models and choose a metric that balances precision and recall.
- Some classifiers do not expose coefficients, so feature explanation must adapt.

Model Comparison:

- Logistic Regression: Accuracy=0.752, Precision=0.367, Recall=0.766, F1=0.497
- Decision Tree: Accuracy=0.759, Precision=0.269, Recall=0.298, F1=0.283
- SVM (RBF): Accuracy=0.810, Precision=0.431, Recall=0.596, F1=0.500
- Random Forest: Accuracy=0.837, Precision=0.444, Recall=0.085, F1=0.143
- Gradient Boosting: Accuracy=0.840, Precision=0.500, Recall=0.191, F1=0.277

Key Insights and Recommendations:

- Best model by F1 score: SVM (RBF) (F1=0.500)
- Use the best model for attrition prediction because it balances false positives and false negatives.
- Consider class balancing techniques, hyperparameter tuning, and cross-validation to improve reliability.
- Tree-based models are useful for feature importance, while SVM performed best for prediction accuracy on this dataset.